

Failure Diagnosis Write-Up

The Failure

During initial testing of the `/classify-ticket` endpoint, the LLM consistently failed to output valid JSON for the `priority` field. Instead of returning one of the enumerated values (`low`, `medium`, `high`, `urgent`), it occasionally output values like `normal` or `critical`, which caused the `Pydantic TicketClassification` model to throw a validation error. This resulted in the API returning a 500-level error with the message "Model output did not match the required schema."

Root Cause Category

Behavior/Instruction Problem & Output Parsing Problem

While the original prompt specified the expected schema, the behavior rules did not strictly constrain the LLM from inventing new values for the `priority` field. The output parser (Pydantic) correctly caught the discrepancy, but the LLM was not properly guided to respect the exact string enums.

The Fix

I updated the `TICKET_CLASSIFICATION_PROMPT` to include a strict **[4. BEHAVIOR RULES]** section, explicitly stating:

1. "Do not invent category values."
2. "Do not invent priority values."

Additionally, I moved the schema definition directly into the **[3. OUTPUT CONTRACT]** section, making the allowed values highly visible. I also implemented structured JSON-mode via `client.chat.completions.create(response_format={"type": "json_object"})` which further forces the LLM to return strict JSON.

Before/After Test Results

Test Set

1. "My laptop is broken and I can't work!"
2. "I need help with my taxes."
3. "I submitted my travel expenses three weeks ago and haven't received the money yet."

Before Fix

- Test 1 Output: `{"category": "workplace", "priority": "critical", ...}` -> **FAILED** (ValidationError)
- Test 2 Output: `{"category": "payroll", "priority": "normal", ...}` -> **FAILED** (ValidationError)
- Test 3 Output: `{"category": "reimbursement", "priority": "medium", ...}` -> **PASSED**

After Fix

- Test 1 Output: {"category": "workplace", "priority": "urgent", ...} -> **PASSED** (Valid Enum)
- Test 2 Output: {"category": "payroll", "priority": "low", ...} -> **PASSED** (Valid Enum)
- Test 3 Output: {"category": "reimbursement", "priority": "medium", ...} -> **PASSED**

The strict behavior rules successfully constrained the LLM to the allowed enum values.